

IP MINI PROJECT PHASE 1

IMAGE PROCESSING DOCUMENTATION

Problem Statement: Art Medium Classification

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Git link: <https://github.com/rekhadhorigol/ip-mini-project-art-medium.git>

1. Introduction:

The objective of this phase is to collect, validate & preprocess the image dataset for Art Medium Classification.

The dataset consists of 125 artwork images belonging to five categories: Oil, Watercolour, Ink, Pastel & Graphite/Pencil, with 25 images in each category.

Since the classification task involves identifying artistic mediums, preserving colour information & textural details is important. Different artistic mediums exhibit distinct visual characteristics, such as brush strokes, pigment distribution, pencil shading, line density & colour blending.

So, preprocessing operations were selected to standardize the images while minimizing the loss of these potentially useful characteristics.

2. Dataset Description:

Art Medium	Number of Images
Pastel	25
Graphite	25
Ink	25
Oil	25
Watercolour	25
Total	125

The images were manually collected from multiple institutional art collections.

The original images varied in dimensions & were primarily stored in JPG format.

3. Dataset Validation:

Before preprocessing, the raw dataset was validated using the MATLAB script `check_dataset.m`.

The following checks were performed:

- Verification of the existence of all five category folders.
- Counting the number of supported image files in each category.
- Checking image readability using `imread`.
- Identifying unreadable or corrupt images.
- Recording the minimum and maximum image dimensions for each category.
- Verifying that each category contained exactly 25 images.

Validation Result:

- Total images: 125
- Number of categories: 5
- Images per category: 25
- Unreadable/corrupt images: 0

The validation process confirmed that the collected dataset was complete & readable.

4. Preprocessing Techniques:

The following operations were implemented using MATLAB.

i) RGB Colour Representation

Purpose: To maintain a consistent three-channel colour representation for all images.

Method: Images containing a single grayscale channel are converted into three identical channels. If an image contains more than three channels, the first three channels are retained.

Reason for selection: Colour is an important characteristic for distinguishing artistic mediums. For example, oil paintings may exhibit rich pigment variations, while watercolour artworks may contain characteristic colour blending & transparency effects.

Maintaining RGB representation ensures that colour information is preserved for subsequent analysis.

Expected output: A three-channel RGB image.

Note: RGB preservation is a representation-standardization step rather than a visual enhancement.

ii) Pixel-Value Normalization

Purpose: To represent image pixel values in a consistent numerical range.

Method: MATLAB's `im2double()` function converts the image into double-precision values, typically mapping 8-bit pixel intensities from $[0, 255]$ to $[0, 1]$.

For an 8-bit image:

$$I_{\text{normalized}} = \frac{I_{\text{original}}}{255}$$

Reason for selection: A normalized numerical representation provides a consistent intensity range for subsequent image-processing operations and potential machine-learning algorithms.

Expected output: A floating-point image with pixel values in the range [0,1].

Storage note: The images are converted back to uint8 before saving as PNG files. So, normalization is performed during computation; the final PNG images are stored as 8-bit images.

iii) Aspect-Ratio-Preserving Resizing

Purpose: To standardize image dimensions without distorting the original artwork.

Method: Each image is resized proportionally using bicubic interpolation.

The scaling factor is calculated as:

$$s = \min\left(\frac{H_t}{H_o}, \frac{W_t}{W_o}\right)$$

Where:

- H_t = Target height
- W_t = Target width
- H_o = Original height
- W_o = Original width
- s = Scaling factor

The new dimensions are calculated using:

$$H_n = \text{round}(sH_o)$$

$$W_n = \text{round}(sW_o)$$

Bicubic interpolation estimates new pixel values using neighbouring pixels & provides smooth resizing.

Reason for selection: The original images have different dimensions & aspect ratios. Directly resizing every image to a square could distort the artwork, potentially altering its proportions & visual characteristics.

Preserving the aspect ratio helps retain the original artwork's shape.

iv) Center Padding

Purpose: To ensure that all images have identical dimensions.

Method: After resizing, each image is placed at the centre of a 256 × 256 canvas. The remaining area is filled with white pixels.

The final dimensions are: 256 X 256 X 3

Reason for selection: Aspect-ratio-preserving resizing does not always produce a square image. Center padding ensures uniform dimensions without stretching or cropping the artwork.

White padding is used to provide a consistent background around the resized image.

5. Why Some Other Techniques Were Not Applied:

Several commonly used image-processing techniques were considered but not applied.

Technique	Reason for Not Applying
Grayscale conversion	Could remove important colour information that distinguishes artistic mediums.
Histogram equalization	Could alter the original intensity & colour distribution of artwork.
Gaussian filtering	Could blur fine brush strokes, pencil details & textures.
Median filtering	Could remove small details & texture patterns that may be useful for classification.
Sharpening	Could artificially enhance edges & alter the original appearance of the artwork.
Cropping	Could remove important portions of the artwork.

The decision was to preserve the original visual characteristics rather than apply transformations that might remove or artificially modify useful information.

6. Input & Output Comparison:

The following comparisons illustrate the preprocessing operations.

Comparison 1: Original Image & Resized Image

- Input: Original artwork with its original dimensions.
- Output: Proportionally resized artwork using bicubic interpolation.
- Observation: The image dimensions are reduced while the original aspect ratio is maintained.

Comparison 2: Resized Image & Padded Image

- Input: Resized artwork.
- Output: Artwork placed at the centre of a 256×256 white canvas.
- Observation: The artwork retains its proportions, while the canvas dimensions become uniform.

Comparison 3: Pixel-Value Normalization

- Input: Original integer pixel representation.
- Output: Floating-point representation with pixel values in $[0,1]$.
- Observation: This is a numerical transformation and does not necessarily produce a visible change in the image.

Screenshots & Outputs:
Input VS Output image eg.:

Import Tool

Import GRA_01.jpg

Variables to be imported:

Name	Size	Class	Value
GRA_01	1712×1389×3	uint8	1712×1389×3 uint8

Preview:

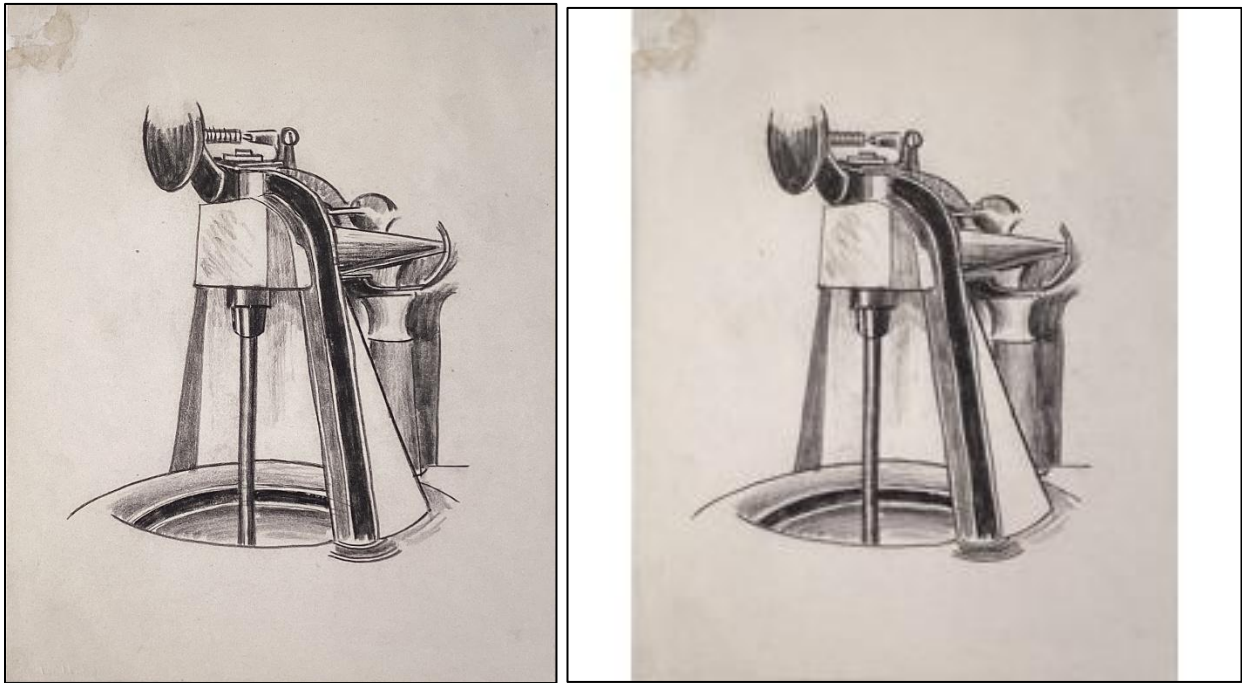
Import Tool

Import GRA_01_processed.png

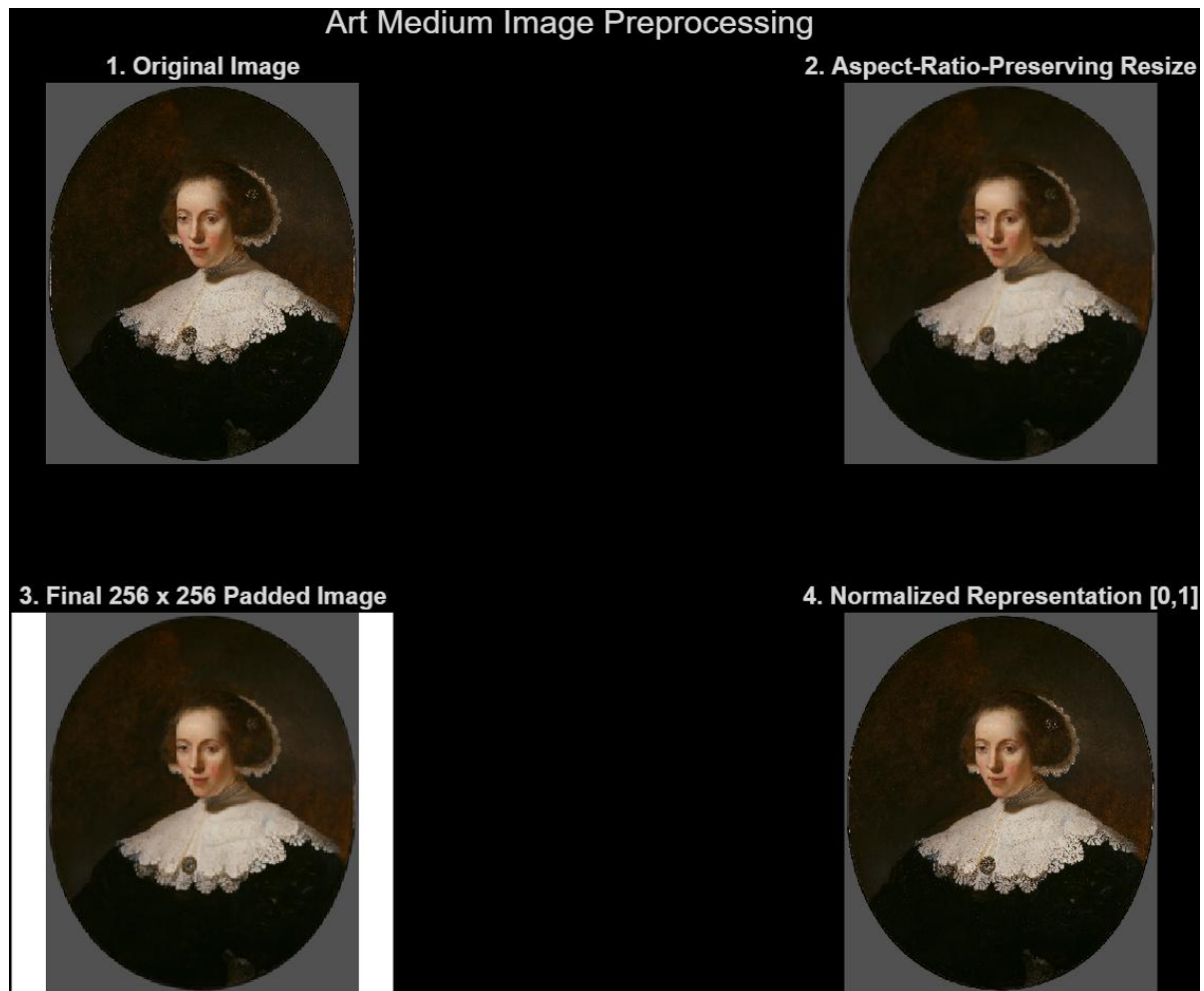
Variables to be imported:

Name	Size	Class	Value
GRA_01_processed	256×256×3	uint8	256×256×3 uint8

Preview:



Detailed stepwise processing eg.: (Output of show_preprocessing_steps.m)



Original data type: uint8

Normalized data type: double

Normalized minimum: 0.0000

Normalized maximum: 1.0000

Resized dimensions: 256 x 210

Final dimensions: 256 x 256

Pixel-value normalization:

Original Pixel Value (8-bit)	Normalized Value [0,1]
0	0.0000
64	0.2510
128	0.5020
192	0.7529
255	1.0000

Explanation: Pixel values are converted from the 8-bit range [0, 255] to a normalized floating-point range [0, 1] using *im2double()*. This normalization is performed during processing & does not produce a separate visible image.

RGB colour preservation: The images are maintained in a three-channel RGB representation so that colour information, which may be an important distinguishing characteristic between different art mediums, is retained for subsequent image analysis & classification.

Output of check_dataset.m:

```
=====
ART MEDIUM DATASET VALIDATION
=====

Dataset location:
/MATLAB Drive/art_medium_dataset/raw

-----
Checking class: OIL
-----

Number of images: 25
Count status: OK (25 images)
Width range : 1536 - 3400 pixels
Height range: 1536 - 3400 pixels

-----
Checking class: WATERCOLOUR
-----

Number of images: 25
Count status: OK (25 images)
Width range : 505 - 1200 pixels
Height range: 645 - 1200 pixels

-----
Checking class: INK
-----

Number of images: 25
Count status: OK (25 images)
Width range : 843 - 4000 pixels
Height range: 646 - 4000 pixels

-----
Checking class: PASTEL
-----

Number of images: 25
Count status: OK (25 images)
Width range : 603 - 4000 pixels
Height range: 450 - 4000 pixels

-----
Checking class: GRAPHITE
-----

Number of images: 25
Count status: OK (25 images)
Width range : 463 - 2792 pixels
Height range: 509 - 3722 pixels

=====
DATASET SUMMARY
=====

Class      ImageCount  MinWidth  MaxWidth  MinHeight  MaxHeight
-----
"oil"      25          1536     3400     1536      3400
"watercolour" 25          505     1200     645      1200
"ink"      25          843     4000     646      4000
"pastel"   25          603     4000     450      4000
"graphite" 25          463     2792     509      3722

Total images: 125
Expected images: 125

Unreadable/corrupt images: NONE

=====
FINAL STATUS
=====
DATASET CHECK: COMPLETE AND VALID
All 5 classes contain exactly 25 readable images.
=====
>>
```

Output of preprocess.m:

```
=====
ART MEDIUM IMAGE PREPROCESSING
=====

Processing: OIL
-----
Images found: 25
Finished: OIL

-----
Processing: WATERCOLOUR
-----
Images found: 25
Finished: WATERCOLOUR

-----
Processing: INK
-----
Images found: 25
Finished: INK
```

```
-----
Processing: PASTEL
-----
Images found: 25
Finished: PASTEL

-----
Processing: GRAPHITE
-----
Images found: 25
Finished: GRAPHITE

=====
PREPROCESSING COMPLETE
=====

Output size: 256 x 256 pixels
Output format: PNG
Color representation: RGB
Interpolation: Bicubic
Aspect ratio: Preserved
Pixel normalization: [0,1] during processing

Processed images are stored in:
/MATLAB Drive/art_medium_dataset/processed

=====
>>
```

7. Final Preprocessing Specifications:

Parameter	Final Specification
Total images processed	125
Number of categories	5
Images per category	25
Output dimensions	256 × 256 pixels
Colour representation	RGB
Interpolation method	Bicubic
Aspect ratio	Preserved
Padding	White, centre-aligned
Processing normalization	[0,1]
Output data type	8-bit unsigned integer
Output format	PNG

All 125 images were successfully processed & saved in their respective category folders.

8. Limitations & Future Work:

i) Dataset source bias:

The dataset was collected from multiple institutional art collections. However, the distribution of image sources is not uniform across all art-medium categories. Consequently, source-related visual characteristics may introduce dataset bias. This limitation should be considered when interpreting future classification results & source-diversified data should be incorporated in subsequent dataset expansion.

ii) Additional image-format support:

The preprocessing script supports JPEG, PNG, BMP & TIFF extensions. However, indexed-colour images require appropriate colour-map handling because imread may return indexed pixel values along with a colour map.

The currently collected dataset consists of JPG images, so this issue does not affect the present dataset. Appropriate indexed-image handling can be incorporated if additional formats are introduced in future dataset expansion.

Future improvements:

- Increase the dataset size.
- Improve source diversity across categories.
- Extend image-format handling.
- Explore suitable image features and classification algorithms in subsequent phases.

9. Conclusion:

The preprocessing process successfully standardized the collected artwork images while preserving their original colour information & aspect ratios.

All 125 images were validated, resized, padded to a common 256×256 dimension & saved in RGB PNG format.

The resulting dataset is organized & prepared for subsequent image analysis, feature extraction & classification stages.